

claude-windows / 45ea3a91-654d-4972-9cd3-65f35db54caf

Metadata

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- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\45ea3a91-654d-4972-9cd3-65f35db54caf\subagents\workflows\wf\_f10e69b6-333\agent-a542dd0e7ec8f354a.jsonl`
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Transcript

1. user (2026-06-15T14:59:53.803Z)

You are reviewing a DRAFT product-overview document for "KhelSutra" ? a local, AI badminton highlight/rally tool.
The draft lives at C:/Users/avidu/Projects/badminton-highlight-indexer/docs/KHELSUTRA_PRODUCT_OVERVIEW.md (Read it). It is dual-audience: (a) a non-technical FIELD ASSOCIATE who will talk to academies, and (b) invited ALPHA/BETA academies/coaches/players who will use the tool and give feedback. It must be PRODUCT-LEVEL (no fine-grained technical detail) and HONEST (no overpromising to real prospective customers).

GROUND TRUTH (from the repo's docs/memory ? what is actually real):

- TODAY the tool genuinely does: upload a long video ? automatic rally detection ? highlight reel (with a "Generated by KhelSutra.guru" watermark) ? a simple summary (rally count + total play time); re-process/download;
private local-first web app (async jobs + chunked upload + highlight download are live & E2E-verified); runs on real 4K GoPro match footage.
- Quality reality (DO NOT put raw numbers in a customer doc): clean single-court footage works very well (~0.93 F1);
busy MULTI-COURT footage is markedly harder (~0.66) and is the explicit moat/focus. The doc should convey this qualitatively ("very good on clean single-court; multi-court is harder and is what we're improving").
- ROADMAP (NOT today): shot detection (smash/drop/serve), deeper per-rally/per-match analysis, per-player history,
"AI assistant coach", more sports (table tennis next), Kannada/Hindi localization.
- THE FLYWHEEL (the headline the owner wants foregrounded): players/coaches can run their OWN videos; coaches can
correct what it found; over time + more videos the model sharpens AND that library unlocks the deeper analysis.
- GUARDRAILS (hard, non-negotiable, must be represented accurately): using a user's video to improve/train the model
is OPT-IN (explicit permission), never automatic; footage involving MINORS is NEVER used for model improvement;
private/local-first ("your video, your data"); pricing is NOT set.
- Capture quality is the gate: fixed camera behind court, full court in frame, landscape, steady, decent light;
handheld/panning/broadcast footage is the failure mode.
- It is NOT line-calling/scoring/umpiring, NOT player biometrics, NOT a coach replacement.

RULES: read-only; you cannot edit. Return STRUCTURED output (consumed programmatically). Quote the draft line/phrase when flagging. Be specific and actionable. Do not invent product capabilities.

You are the editor. Below are 4 lens reviews of the draft (JSON). Read the draft once yourself, then produce a single

DEDUPED, PRIORITISED edit list I will apply before generating the shareable PDF. Resolve any conflicts between lenses

(state the call). Separate BLOCKING edits (honesty/consent ? must fix before this goes to a customer) from

IMPROVEMENTS. For each edit give the exact target text and the exact replacement so I can apply it mechanically.

Also give a 2-3 sentence overall readiness verdict.

REVIEWS: [{"lens": "overpromise", "verdict": "The draft is unusually disciplined for a customer-facing doc: it has an explicit \"Read this honestly\" roadmap callout, a strong \"What KhelSutra is not\" section, and crisp guardrail language (opt-in, minors-never-used, local-first, pricing-not-set) that matches ground truth. The honesty failures are concentrated in two places that an alpha customer would absolutely read as promises the product does not keep today: (1) completeness/accuracy language (\"finds every rally\") and, more seriously, (2) repeated phrasing that implies busy multi-court footage is already solved ? which directly contradicts the doc's own weak-spot disclosures and the ground-truth ~0.66 reality. The flywheel section is also written in the present tense as if per-user model sharpening is live, when it is roadmap. Fix those and the doc cleanly under-promises.\""}, {"must_fix": [{"quote": "***KhelSutra turns ordinary badminton video into useful pieces ? automatically.** Point a phone or\\nGoPro at the court, record a session, and KhelSutra finds every rally\", \"problem\": \"\"finds every rally\" is a completeness guarantee the product does not keep. Ground truth is ~0.93 F1 on clean single-court and ~0.66 on busy multi-court ? it misses and over-calls rallies, especially in exactly the multi-court scene the doc elsewhere calls the hard case. An alpha academy will count missed/spurious rallies in their first reel and read this as a broken promise.\""}, {"fix": "Softening to an automatic-detection claim without the completeness guarantee, e.g. \"...record a session, and KhelSutra automatically finds the rallies, stitches a highlight reel...\" Drop \"every.\"\""}], [{"quote": "the underlying model gets better at\\n*your* courts, *your* lighting, *your* players, and the busy multi-court halls that trip up everything\\nelse. It improves with use instead of staying frozen.\""}, {"problem": "Two problems. (a) \"the busy multi-court halls that trip up everything else\" implies KhelSutra is NOT tripped up by multi-court ? i.e. multi-court is solved. This contradicts ground truth (~0.66, the explicit moat/focus) AND the doc's own later admissions (the \"not\" section and the accuracy FAQ both call multi-court the hard/weak case). (b) The whole passage is present tense (\"gets better,\" \"improves with use\"), implying per-user model sharpening is a live feature. Per ground truth/current-state the per-user/personalization flywheel is roadmap (per-user model store is persisted but not yet read on the served path); the headline is wanted but must be framed as direction/design, not a feature working today.\""}, {"fix": "Reframe to direction + remove the multi-court-solved implication, e.g. \"This is the model KhelSutra is built around: as you and your players run your own footage through it ? and as coaches confirm or correct what it found ? that growing, labelled library is what we use to sharpen the tool for your courts, your lighting, and the busy multi-court halls that are the hardest case. It's designed to improve with use rather than stay frozen.\" Keep it future/design-tense and stop asserting multi-court already beats \"everything else.\"\""}], [{"quote": "And most academy halls run **several courts at once**, so a single recording is a busy, noisy scene.\\nOrdinary video tools choke on that. **Pulling the real rallies out of that footage ? automatically ?\\nis exactly what KhelSutra is built to do.**\""}, {"problem": "Placed immediately after describing busy multi-court scenes and bolded, a prospect reads this as \"KhelSutra reliably pulls rallies out of busy multi-court footage today.\" That is the exact case ground truth flags as markedly harder (~0.66) and still being improved. \"built to do\" is mission framing, but the adjacency makes it a de facto current-capability claim about the hardest scenario.\""}, {"fix": "Make it a mission/focus statement, not an implied solved-today claim, e.g. \"...Ordinary video tools choke on that. Pulling the real rallies out of busy, multi-court footage is the hard problem KhelSutra is built to crack ? it works very well on clean single-court recordings today, and multi-court is exactly what we're sharpening.\" This also pre-aligns the intro with the honest FAQ answer later.\""}], [{"quote": "with a small\\n*\"Generated by KhelSutra.guru\\n\"* mark.\""}, {"problem": "The draft quotes the watermark text verbatim in quotation marks, but the actual configured watermark in config.json is \"Generated by Khelsutra.guru\" (lowercase 's'). A field associate showing a real reel will display text that doesn't match the doc's quote. Either the doc or the config is wrong; the doc should not assert a verbatim string it doesn't match.\""}, {"fix": "Either correct the quote to match the shipped watermark (\"Generated by Khelsutra.guru\") or, better for a non-technical doc, describe it without quoting exact casing: \"...with a small KhelSutra watermark.\" Flag to the owner to reconcile config.json casing if the capital-S branding is intended.\""}], [{"quote": "***\"Does it work for doubles and tournaments?\\n***\\nYes for finding rallies. Crowded multi-court tournament halls are the hard case we're specifically working on.\""}, {"problem": "\"Yes for finding rallies\" is an"}]

unqualified yes that a prospect generalizes to all doubles/tournament footage, then the next sentence concedes crowded multi-court is the hard case ? which is most tournament footage. The unqualified \"Yes\" slightly overpromises before the caveat lands.\", \"fix\": \"Tie the yes to the capture conditions, e.g. \\\"It finds rallies in doubles too. Tournament footage varies: a clean, fixed full-court recording works well; crowded multi-court tournament halls are the hard case we're specifically working on.\\\"\", {\"quote\": \"**Automatic rally detection** ? it finds where each rally starts and ends, on its own.\", \"problem\": \"\\\"finds where each rally starts and ends\\\" again leans on completeness/precision (\\\"each,\\\" exact start/end). Detection and boundaries are imperfect (the doc's own padding/trim behavior and ~0.66 multi-court reality). Minor relative to the must-fixes, but it compounds the \\\"every rally\\\" completeness framing.\", \"fix\": \"Soften to \\\"Automatic rally detection ? it finds the rallies and where they start and end, on its own.\\\" Avoid \\\"each\\\" as a per-rally completeness claim.\"}], \"nice_to_have\": [\"The accuracy FAQ answer (\\\"Very good on a clean, single-court recording. A busy multi-court hall is harder ? and is exactly what we're improving. The best answer is to show a sample reel rather than quote a number.\\\") is the gold-standard honest framing in this doc. Consider mirroring its exact qualitative split (\\\"very good on clean single-court; multi-court is harder and is what we're improving\\\") into the intro and the flywheel section so the whole doc speaks with one honest voice rather than over-claiming up top and correcting in the FAQ.\", \"The \\\"Read this honestly\\\" callout (lines 104-106) is excellent under-promise discipline. A parallel one-line honesty nudge inside the flywheel section (\\\"the per-user sharpening described here is the design we're building toward; what runs today is rally detection + reels + summary\\\") would close the last present-tense gap without weakening the headline.\", \"Consider stating once, plainly, that the tool can both miss real rallies and include non-rallies (false positives/negatives) so early users frame their feedback as expected-and-wanted rather than as the product being broken ? this also reinforces the correction-loop ask.\"], \"praise\": [\"The dedicated \\\"What KhelSutra is not\\\" section is exactly right for an alpha audience and accurately represents the hard guardrails: not line-calling/scoring/umpiring, not a coach replacement, shot-detection/AI-coach explicitly roadmap-not-today, accuracy depends on capture quality, pricing not set.\", \"Guardrail language is faithful to ground truth and crisp: opt-in (\\\"nothing is used to train anything without your explicit permission\\\"), \\\"Sessions involving minors are never used for model improvement,\\\" \\\"private and local-first... your video, your data,\\\" and \\\"Pricing isn't set.\\\" These are the non-negotiables and they are represented accurately.\", \"The roadmap section is correctly labelled as direction with the bolded \\\"Read this honestly\\\" callout stating plainly that those items are \\\"the direction, not today's buttons\\\" and re-anchoring on what works today ? this is the right under-promise posture.\", \"The capture-quality guidance (fixed camera behind court, full court in frame, landscape, steady, decent light; handheld/panning/broadcast as the failure mode) matches ground truth precisely and is framed as honest expectation-setting, not a hidden disclaimer.\", \"The accuracy FAQ's \\\"show a sample reel rather than quote a number\\\" is precisely the right instinct given the instruction to keep raw numbers out of a customer doc.\"}], {\"lens\": \"consent-safety\", \"verdict\": \"Through the consent/privacy/safety lens this draft is close but has one disqualifying problem: it repeatedly markets the product as \\\"private and local-first\\\" / \\\"your video stays yours,\\\" yet the actual alpha/beta product uploads video to a server and sends it to Google's Gemini cloud API for processing (CLAUDE.md line 3; HOSTING_PLAN.md alpha + beta topologies). The repo's own PERSONALIZATION_PLAN.md (line 145) explicitly forbids advertising data-locality until on-device ships ? so this is not a nuance, it is a guardrail the doc breaks in front of real prospective customers. The opt-in and minors claims are directionally right and prominent, but the opt-in claim is overstated relative to the planned free-tier \\\"consent baked into the terms\\\" pooling, and the minors rule is stated more narrowly here than in the field flyer. Fix the locality framing and tighten the opt-in/minors wording before this goes to any academy.\", \"must_fix\": [{\"quote\": \"The product is built **private and local-first**: your video, your data, your control.\", \"problem\": \"FALSE for the current product and a direct violation of an internal guardrail. CLAUDE.md (line 3) states the product runs \\\"with paid Gemini as the cloud AI\\\"; HOSTING_PLAN.md shows alpha sends video from the box to the Gemini API and beta uploads browser->GCS and processes Gemini-path jobs \\\"entirely in-cloud.\\\" PERSONALIZATION_PLAN.md line 145-146 explicitly says: \\\"Privacy deferral - do not advertise 'data never leaves your machine' until on-device ships; near-term pooling carries the full cloud (DPA/consent/minors) posture.\\\" On-device/local is a FUTURE end-state (STORAGE_SHARING_MODEL.md 7d), not today. A coach reading 'local-first / your video stays yours' will reasonably believe footage never leaves their premises - which is untrue.\", \"fix\": \"Drop 'local-first' and any 'data never leaves your machine' implication for the current product. State the truth plainly, e.g.: 'Your video is private to you and is processed securely for you - it is not shared or published. Today, processing happens on our secure servers (and, for some steps, a trusted cloud AI provider); we are working toward an on-device option where your footage never leaves your own machine.' Keep 'your video, your data, your control' only if paired with the accurate processing description.\"}], {\"quote\": \"**\\\"Is my video private? What

happens to it?"**\nIt's built private and local-first ? your video stays yours.", "problem": "Same false locality claim repeated in the customer-facing FAQ, which is the exact place a prospect will scrutinize the privacy promise. 'Private and local-first' cannot be stated for a product that uploads to a server and calls Google's Gemini cloud API. This is the highest-stakes line in the doc and is currently inaccurate.", "fix": "Rewrite to describe the real data flow honestly: 'Your video is private to you - we never publish it and we don't use it to improve the model without your explicit opt-in. To produce your reel, your video is processed on our secure infrastructure (which today includes a trusted cloud AI step); footage involving minors is never used to improve the model. An on-device option (your footage never leaving your machine) is on the roadmap.' Avoid the words 'local-first' until on-device ships."}, {"quote": "Using your video to improve the model is **opt-in** ? nothing is used to train\nanything without your explicit permission.", "problem": "Overstated relative to the actual plan, which risks an honesty problem the moment a free tier launches. PERSONALIZATION_PLAN.md (decision 5, line 57) states: \"a free tier is likely and free-tier videos ARE pooled into the baseline (consent baked into the free terms ... minors-exclusion still enforced)\". 'Consent baked into the free terms' is bundled/terms-of-service consent, NOT the affirmative, separable 'explicit permission' the doc promises. As written, the doc could later read as a broken promise to a free-tier user. (Pricing isn't set, so this is a near-future contradiction, but it is in the authoritative plan and should not be foreclosed by an absolute promise now.)", "fix": "Keep the strong opt-in promise but scope it to what is guaranteed: 'We will never use your video to improve the model without your clear, informed agreement, and you can decline. If we ever offer a free tier where contributing footage is part of the deal, that will be stated upfront and in plain language - and footage involving minors is never used either way.' This stays honest whether or not free-tier pooling ships."}], "should_fix": [{"quote": "***Sessions involving minors\nare never used for model improvement.**", "problem": "The minors rule is correct but stated more NARROWLY than the rest of the company's stated posture, and a customer-facing doc should not under-state it. The field-associate flyer (PRODUCT_FIELD_ASSOCIATE_FLYER.md line 76-78) commits that 'sessions involving minors are never recorded for our use, period.' STORAGE_SHARING_MODEL.md guardrails (line 153, 182) add 'minors never on public profiles' as a separate, stricter rule. Restricting the public promise to only 'model improvement' could be read as 'minors' footage is fine to publish/share, just not to train on' - which contradicts the stricter internal posture and is a safety-sensitive gap for a youth-sport product.", "fix": "Broaden the minors line to cover both training AND publishing, matching the internal posture: 'Footage involving minors is never used to improve the model and is never published or put on any public profile.' Apply the same broadened wording to the FAQ answer ('footage involving minors is never used')."}, {"quote": "We don't publish anyone's footage.", "problem": "Good and accurate today, but the roadmap (STORAGE_SHARING_MODEL.md 7c social layer: profiles, share links, reel galleries, public/unlisted/private visibility) anticipates user-initiated sharing. An absolute 'we don't publish anyone's footage' could be read as 'sharing will never exist,' setting up a future surprise. Minor, since it is true now.", "fix": "Tighten to make clear the no-publish promise is about KhelSutra never publishing on its own: 'We never publish your footage - you decide what, if anything, you share.' This stays true if user-controlled sharing ships later."}], {"quote": "the underlying model gets better at\n*your* courts, *your* lighting, *your* players", "problem": "Consent-adjacent: 'your players' implies the system learns from identifiable individuals' footage, while the product is explicitly NOT player biometrics/identity (CLAUDE.md line 52-53: gait/biometrics strictly future, GDPR Art. 9). For a youth-heavy academy audience, 'gets better at your players' can be misread as per-person tracking/identification. Worth disambiguating so it is not read as a biometric claim.", "fix": "Reword to make clear this is about scene/footage adaptation, not identifying people: 'gets better at your courts, your lighting, and the kind of play you record.' Drop 'your players' or qualify it explicitly as not player identification."}], "nice_to_have": ["Add a one-line, plain-language data-handling summary near the privacy paragraph that a non-technical field associate can repeat verbatim and stand behind (e.g., 'We process your video to make your reel; we don't publish it; we don't train on it without your yes; we never train on minors' footage'). The field associate is non-technical and will be asked these questions live - giving them an exact, accurate script reduces the risk of them ad-libbing an overpromise.", "Consider stating retention plainly: per STORAGE_SHARING_MODEL.md (line 29) raw uploads are kept only through processing + a correction window, then auto-deleted. A short 'we don't keep your raw video longer than needed to make and let you correct your reel' is a strong, honest privacy positive the draft currently omits - though confirm it is actually implemented before claiming it.", "If the team wants to keep any locality language at all, restrict it to 'private to you' (access control) rather than 'local' (physical location), since the former is defensible today and the latter is not."], "praise": ["The dedicated privacy paragraph ('Your data stays yours.') and the FAQ both surface opt-in, minors-exclusion, and no-publish prominently - privacy is not buried, which is"]

exactly right for a customer doc.", "The minors-exclusion rule appears in BOTH the body and the FAQ - good redundancy for the single most safety-sensitive claim in a youth-sport context.", "The 'What KhelSutra is not' section and the 'Read this honestly' callout set accurate expectations (not line-calling, not a coach replacement, roadmap-vs-today separation, pricing not set) - this honesty discipline is the right posture and should be preserved.", "The opt-in framing leads with 'nothing is used without your explicit permission' rather than burying it in terms - the instinct is correct; it just needs scoping so a future free tier doesn't break it."]}, {"lens": "grounding", "verdict": "From a grounding/factual-accuracy lens, the draft is largely honest and well-aligned with the repo: the \"what it does today\" core loop (upload long video ? automatic rally detection ? highlight reel + simple summary ? re-use/download), the capture rules, the multi-court framing, the opt-in/minors/local-first guardrails, and the roadmap-not-today disclaimers all match the ground truth. Two real factual errors must be fixed before this goes to prospective customers: (1) the watermark text is mis-cased (\"KhelSutra.guru\" vs the actually-shipped \"Khelsutra.guru\"), and (2) the flywheel section overstates the mechanism ? today nothing auto-improves and there is no per-user training; \"it learns from your videos / gets sharper at your courts\" describes a roadmap aspiration, not a shipped capability, and reads as overpromising to alpha users. A few framing/consistency tightenings round it out.", "must_fix": [{"quote": "with a small *\"Generated by KhelSutra.guru\"* mark.", "problem": "Watermark text is mis-cased. The actually-shipped watermark string is \"Generated by Khelsutra.guru\" (lowercase 's' in Khelsutra) ? confirmed in config.json line 79, backend/config/models.py:178 (default `text`), README.md lines 28/120/197, and tests/test_watermark.py:25. A customer who looks at a reel will see lowercase; quoting the wrong casing in the customer-facing doc is a factual mismatch about what the product literally stamps on output.", "fix": "Change to the exact shipped string: *\"Generated by Khelsutra.guru\"* (lowercase 's'). Use this casing anywhere the watermark text is quoted verbatim. (The product NAME is styled \"KhelSutra\"/\"KhelSutra Guru\" per ADR-009 ? that's fine; only the literal watermark string must match what's burned in.)"}], {"quote": "***Your videos make _your_ tool sharper.** As you and your players run more of your own footage through KhelSutra ? and as coaches confirm or correct what it found ? the underlying model gets better at *your* courts, *your* lighting, *your* players, and the busy multi-court halls that trip up everything else. It improves with use instead of staying frozen.", "problem": "This overstates the shipped mechanism and is not grounded. Per the guardrails (CLAUDE.md) and the owned-model roadmap (NEXT_STEPS.md / OWNED_MODEL_IMPLEMENTATION_PLAN.md), there is NO automatic, no per-user, and no live model improvement today: training is a future, owner-gated, opt-in, batch process on a consented corpus ? the model does NOT \"get better at your courts\" as you upload, and there is no per-user/per-academy fine-tuning at all (the moat is one shared consented dataset, not personalized models). 'gets better at *your* courts, *your* lighting, *your* players' implies a personalized self-tuning loop that does not exist. To alpha academies who will actually use the tool, this reads as a promise the product can't keep this year.", "fix": "Reframe as collective + future + opt-in, not personal + live. e.g.: 'The footage that real academies contribute (with permission) is what lets us improve KhelSutra over time ? especially on busy multi-court halls, the hard case. Your corrections tell us where it got a rally wrong. Today the tool doesn't retrain itself as you use it; improvements ship in new versions as the contributed library grows.' Drop 'your courts / your lighting / your players' (implies a per-user model) and 'It improves with use' (implies a live self-tuning loop)."}], {"should_fix": [{"quote": "***Personalised insights** ? an **AI assistant coach** that supports (never replaces) the real coach, surfacing what to work on next.", "problem": "'Personalised insights' for an 'AI assistant coach' again leans into per-user personalization, which compounds the flywheel overstatement and is the furthest-out, least-defined roadmap item. It's correctly inside the 'Where it's going' section with a 'direction not today's buttons' disclaimer, so it isn't a hard must-fix ? but the word 'Personalised' echoes the per-user-model implication flagged above.", "fix": "Soften to avoid the personalization claim, e.g. 'Coaching insights ? an AI assistant coach that supports (never replaces) the real coach, surfacing what to work on next.' Keep the existing honest disclaimer in the section."}], {"quote": "the underlying model gets better", "problem": "Minor altitude/technical-leak issue for a product doc: 'the underlying model' exposes implementation framing (there is a model) that the rest of the doc otherwise keeps product-level. Combined with the must-fix above, it nudges readers toward expecting a live learning system.", "fix": "Prefer product-level phrasing like 'KhelSutra gets better over time (in new versions)' rather than 'the underlying model gets better'."}], {"quote": "***Re-use & download** ? process once, come back to it, download the reel.", "problem": "Grounded but worth a precision check: re-processing the SAME file is a cache hit / re-ingest (UPSERT, results replaced not merged ? README Q11, CODE_MAP §db). 'process once, come back to it' is fine as written for the customer altitude; just ensure field associates don't verbally promise that re-running keeps prior versions side-by-side (it replaces). No doc text change strictly required.", "fix": "Optional: leave the line as-is (it's honest at product altitude). If you want to pre-empt confusion, add to the FAQ: 'Re-processing the same video

replaces the earlier result rather than keeping both.'"},"quote":"It's built private and local-first ? your video stays yours.", "problem":"Consistency check vs COMMERCIALIZATION.md: the production serving path is a hosted GPU VM with a paid Gemini call (paid tier = no training/review), and the hosting docs (HOSTING_PLAN.md, khelsutra.guru) describe a cloud-served app. 'Local-first' is the accurate product-philosophy term used across the repo (README, ADR-009) and is fine, but for prospective customers 'your video stays yours' should not imply the video never leaves their premises during processing.", "fix":"Keep 'private' and 'your video, your data', but make sure the privacy FAQ doesn't read as 'your video never leaves your device.' Consider: 'We treat your footage as private ? it isn't published, isn't used to train anything without your explicit opt-in, and footage involving minors is never used for model improvement.' (Matches the guardrails without implying zero cloud processing.)"}], "nice_to_have":["The closing line 'Field associates: the conversation script and visit rules live in the separate Field Visit Playbook' is accurate and correctly points to FIELD_VISIT_PLAYBOOK.md ? good cross-doc hygiene. Consider also naming PRODUCT_FIELD_ASSOCIATE_FLYER.md so the recruiting vs working-playbook split is clear.", "The capture rules in this doc ('behind the court, slightly elevated, whole court in frame, keep it still, landscape, reasonable lighting') match the Field Visit Playbook Phase-2 checklist and VIDEO_RECORDING_GUIDELINES. The Playbook also specifies '1080p/60fps' ? if you want full consistency you could mention a resolution/framerate hint, but omitting it is appropriate for this audience.", "The 'researching, not selling' stance is implied by 'invited alpha/beta' and 'pricing isn't set,' which is consistent with the Field docs' explicit 'You are gathering feedback, not selling.' Fine as-is for a product overview; the harder no-selling rule correctly lives in the Field docs.", "'~4 weeks' focus window matches the field engagement duration (~3?4 weeks) in the flyer/playbook ? consistent.", "praise":["The 'Read this honestly' callout (lines 104-106) and the entire 'What KhelSutra is not' section are exemplary for an honest customer doc ? they correctly scope shot detection / AI-coach analytics as roadmap-not-today, name capture quality as the gate, and disclaim line-calling/scoring/coach-replacement, all matching the ground truth.", "The guardrails are represented accurately: opt-in training ('nothing is used to train anything without your explicit permission'), 'Sessions involving minors are never used for model improvement,' and 'private and local-first' all match the non-negotiable guardrails in CLAUDE.md/COMMERCIALIZATION.md C14.", "The accuracy framing is honest and matches the quality reality without leaking raw numbers: 'Very good on a clean, single-court recording. A busy multi-court hall is harder ? and is exactly what we're improving' correctly conveys the ~0.93 vs ~0.66 split qualitatively and foregrounds multi-court as the moat/focus.", "The 'today vs roadmap' split is factually correct: rally detection + highlight reel + simple summary (rally count + play time) + re-process/download are exactly the live, E2E-verified capabilities per NEXT_STEPS (async jobs #87, chunked upload/download #99); shot detection, per-player history, AI coach, table-tennis-next, and Kannada/Hindi are all genuinely roadmap items.", "Product naming is consistent with ADR-009: 'KhelSutra' / 'khelsutra.guru' matches the ratified product name and domain across the repo."}], "lens":"clarity", "verdict":"This is a strong, honest, well-structured draft that mostly nails the dual-audience brief: a coach can read it top-to-bottom and understand what they get today, and an associate gets a usable mental model. The flywheel is foregrounded well and the guardrails are accurate. The main gaps are usability-for-the-alpha-USER (the doc explains the tool but never tells a user HOW to actually use it ? where to upload, what a \"job\" looks like, how long to wait, how to give feedback) and a few honesty/precision nits (a watermark-text mismatch with the real default, and an over-strong \"every rally\" claim that the multi-court reality contradicts). It also slightly overshoots length/repetition. Tightening those would make it both more honest and more actionable for both readers.", "must_fix":[{"0":"f", "quote":"with a small *\"Generated by KhelSutra.guru\"* mark.", "problem":"Mismatch with the actual product. The shipped default watermark text in backend/config/models.py:178 is \"Generated by Khelsutra.guru\" (lowercase 's' in 'Khelsutra'). The doc capitalizes it as \"KhelSutra.guru\". An alpha user who notices the reel says something different from the doc loses trust; an associate may quote it wrong. Either fix the doc to match the code, or fix the code default ? but they must agree.", "fix":"Quote it exactly as it renders today: *\"Generated by Khelsutra.guru\"* ? or, if the brand casing should be KhelSutra, flag to the owner that the config default text needs updating so doc and product match. Also appears in the \$What it does today bullet (line 45)."}], "quote":"KhelSutra finds every rally", "problem":"Overpromise that the ground truth explicitly contradicts and that the doc itself walks back later. On busy multi-court footage the tool is markedly weaker (the stated moat/weak spot), so it does NOT find 'every' rally. A real prospective customer reading 'finds every rally' in the very first line, then 'multi-court is the hard case' on page 2, will read that as a contradiction ? and on clean footage 'every' is still an absolute that one missed rally falsifies in a demo.", "fix":"Softens the absolute, e.g. 'finds the rallies' or 'automatically finds the rallies and stitches a highlight reel'. Reserve quality nuance for the capture and FAQ sections where it already lives."}], "should_fix":[{"quote":"This is what's working right now, in the private web app

invited users can try:","problem":"Usability gap for the alpha-USER audience. The brief requires that an alpha user can read this and 'know how to use the tool well.' The doc describes WHAT the tool does but never the HOW: there's no mention of where the app is (a URL/login), that uploads are chunked and large files take time, that processing is an async job that runs in the background (you don't wait staring at it), roughly how long a session takes, or where the reel/summary appear when done. A coach left at 'upload a video' won't know what success looks like.", "fix":"Add a short 'How to use it (alpha)' subsection or 3-4 line numbered walkthrough: e.g. 1) open the app at <link> / we set you up, 2) upload your session video (large files upload in pieces, can take a while on Indian uplinks), 3) it processes in the background ? you can close the tab and come back, 4) when done you get the reel + summary to watch and download, 5) re-process anytime. Keep it product-level, no internals."}, {"quote":"## What we're focused on over the next ~4 weeks", "problem":"Mostly addressed to the team/associate, not the alpha user, and it's the weakest fit for the 'know what feedback to give' goal. The brief asks that an alpha user knows 'what feedback to give' ? but the doc never tells a user HOW or WHAT to feed back. The roadmap/focus sections tell them what's coming, not what the team wants from them this month.", "fix":"Add a tight 'What feedback helps us most' box for alpha users (3-5 bullets): tell us when it missed a real rally or grabbed a non-rally, whether the reel is the right length/order, what summary numbers you wish you had, and your real capture conditions (court count, camera, lighting). Pair it with how to send feedback (WhatsApp/the associate/a form). This converts 'co-builders' from a slogan into an action."}, {"quote":"That same growing library of real, labelled footage is also what unlocks the **deeper features below** ?\n\nwhich is why early users aren't just customers, they're co-builders. Every session you contribute makes\nthe next version smarter.", "problem":"Repetition that inflates length without adding information. The flywheel idea ('improves with use', 'co-builders', 'every session makes the next version smarter') is stated three times across this section, the 'Where it's going' callout (line 106), and the FAQ 'Will it get better?'. For a ramp-up doc the brief flags length as a constraint; the message lands once.", "fix":"Keep the crisp first statement (the 'Your videos make your tool sharper' paragraph) and trim one of the restatements ? e.g. shorten this paragraph to a single sentence linking the library to the roadmap, and let the FAQ carry the 'will it improve' answer."}, {"quote":"the underlying model gets better at\n*your* courts, *your* lighting, *your* players", "problem":"Mild honesty tension with the opt-in guardrail two paragraphs later, for a careful reader. 'Gets better at YOUR courts/players' reads like automatic per-customer learning, but the very next paragraph says improvement is opt-in and minors are never used. A coach could ask 'so does it learn from my videos or not?' The two claims are reconcilable but sit awkwardly.", "fix":"Tie the conditionality into the same breath: e.g. 'when you choose to share footage (opt-in, never automatic), the model gets better at your courts, your lighting, your players.' This makes the consent and the benefit one coherent statement."}, {"quote":"A one-page capture checklist is provided to academies in the trial.", "problem":"Forward-reference to an artifact the reader can't see, with no path to it. Both audiences benefit from knowing where it is. For the associate it's a deliverable they hand over; for the user it's the single most quality-determining thing. As written it's a promise with no handle.", "fix":"Note who provides it and when (e.g. 'your field contact / the associate gives you the one-page capture checklist at setup'), so it's actionable rather than a dangling reference."}], "nice_to_have":["The 'What KhelSutra is not' and the FAQ overlap heavily (line-calling/scoring, capture-quality dependence, pricing-not-set, shot-detection-is-roadmap all appear in both). Consider trimming the 'not' list to the 2-3 points the FAQ doesn't cover, or merging ? this is the easiest length win.", "The opening promises 'No special hardware' then the very next sections require a tripod, elevated position, landscape, and decent light. Not a contradiction, but a first-time coach reading 'no special hardware' then a four-bullet capture discipline may feel mild whiplash. Consider 'No special hardware ? just a phone or GoPro on a tripod' so the constraint is set from line one.", "Add a one-line 'what you get back, concretely' so a coach pictures the output: e.g. 'one MP4 highlight reel plus a short summary (rally count + total play time)' ? the word 'reel' is used a lot but the deliverable format is never named.", "The doc never states there is no English/Kannada/Hindi UI choice TODAY (localization is roadmap). A Bengaluru coach might assume Kannada is available now because it's listed. The roadmap callout covers it, but a single 'today the app is in English' would prevent a wrong assumption.", "Consider a one-line 'how to reach us / your contact' footer for alpha users ? the doc tells associates where their playbook is but gives the USER no support channel."], "praise":["The TODAY-vs-ROADMAP separation is excellent and honest: the bolded 'Read this honestly' callout (line 104-106) and the dedicated 'What KhelSutra is not' section are exactly the honesty rails the brief demands, and they're written in plain coach-friendly language.", "Guardrails are represented accurately and prominently: opt-in (never automatic), minors-never-used, no-publishing, private/local-first, and pricing-not-set all appear, and the 'Your data stays yours' framing is the right register for a non-technical owner.", "Strong product-level discipline ? no model names, no F1 numbers, no pipeline internals leak in. The 'How accurate is it?' FAQ answer ('show a sample reel rather than quote a number')

is a smart, honest move that matches the real quality reality without exposing figures.", "The 'capture matters' section is genuinely useful to both audiences and correctly identifies handheld/panning/broadcast as the failure mode ? this is the single highest-leverage thing for a new user and it's given real estate.", "Good audience signposting throughout (who it's for, Bengaluru-first focus, the footer pointing associates to the separate Field Visit Playbook) ? it complements rather than duplicates the existing PRODUCT_FIELD_ASSOCIATE_FLYER.md and FIELD_VISIT_PLAYBOOK.md.", "The FAQ covers the real academy questions well: accuracy, hardware, privacy, scoring/line-calling, doubles/tournaments, will-it-improve, sports, and how-to-join ? that's the right set for a Bengaluru academy conversation."]]]

2. assistant (2026-06-15T14:59:57.485Z)

I'll start by reading the draft document myself to ground my review, then reconcile the four lens reviews against what I find.

3. user (2026-06-15T14:59:58.363Z)

```
1      # KhelSutra ? Product Overview & Early-User Guide
2
3      *What it does today, how to get the best out of it, and where it's going.*
4      *For field associates and for invited alpha / beta academies, coaches, and players.*
5
6      ---
7
8      ## In one line
9
10     **KhelSutra turns ordinary badminton video into useful pieces ? automatically.** Point a
phone or
11     GoPro at the court, record a session, and KhelSutra finds every rally, stitches a
**highlight reel**,
12     and gives you a **simple summary** (how many rallies, how much real play time). No
special hardware,
13     no editing, no manual scrubbing.
14
15     ---
16
17     ## The problem we're solving
18
19     Academies and serious players already record a lot of video ? and then almost never
watch it. Hours
20     of footage pile up on phones and memory cards because cutting it into useful clips by
hand is tedious
21     and nobody has the time. The good rallies, the moments worth reviewing, stay buried.
22
23     And most academy halls run **several courts at once**, so a single recording is a busy,
noisy scene.
24     Ordinary video tools choke on that. **Pulling the real rallies out of that footage ?
automatically ?
25     is exactly what KhelSutra is built to do.**
26
27     ---
28
29     ## Who it's for
30
31     - **Coaching academies & coaches** ? turn raw session footage into review-ready clips
and reels for
32     teaching, without an editor.
33     - **Serious players** ? get clips of your own rallies to review, keep, and share.
34     - Today's focus is **badminton academies in Bengaluru** ? that's who we're building with
first.
35
36     ---
37
38     ## What it does **today**
39
```

40 This is what's working right now, in the private web app invited users can try:

41

42 - ****Upload a full session or match video**** (from a phone on a tripod, or a GoPro). Long videos are fine.

43 - ****Automatic rally detection**** ? it finds where each rally starts and ends, on its own.

44 - ****Highlight reel**** ? the rallies stitched into one clip you can watch and share, with a small

45 **"Generated by KhelSutra.guru"** mark.

46 - ****A simple summary**** ? how many rallies were found and how much actual play time there was.

47 - ****Re-use & download**** ? process once, come back to it, download the reel.

48

49 That's the core loop: ****record ? upload ? get back a reel + summary.**** No editing skill needed.

50

51 ---

52

53 **## How to get the best out of it (capture matters)**

54

55 KhelSutra works best on a ****steady, full-court recording**** ? the same way a coach would set a camera

56 to review play. A few simple habits make a big difference to the results:

57

58 - ****Put the camera behind the court, slightly elevated****, with the ****whole court in frame****.

59 - ****Keep it still**** ? a fixed tripod beats a handheld or panning shot every time.

60 - ****Shoot landscape****, with reasonable lighting.

61 - ****Record the whole session**** and upload it as-is ? KhelSutra does the cutting.

62

63 Footage that fights these habits (shaky handheld, heavy zooming/panning, broadcast-style cuts, very

64 dark halls) is the ****hardest case**** and where results suffer most. When in doubt: fixed camera, full

65 court, don't touch it.

66

67 > A one-page capture checklist is provided to academies in the trial.

68

69 ---

70

71 **## The part that makes KhelSutra different: **it learns from your videos****

72

73 This is the heart of the product, and what sets it apart from a generic "AI clipper":

74

75 ****Your videos make _your_ tool sharper.**** As you and your players run more of your own footage through

76 KhelSutra ? and as coaches confirm or correct what it found ? the underlying model gets better at

77 ***your* courts, *your* lighting, *your* players, and the busy multi-court halls that trip up everything**

78 else. It improves with use instead of staying frozen.

79

80 That same growing library of real, labelled footage is also what unlocks the ****deeper features below**** ?

81 which is why early users aren't just customers, they're co-builders. Every session you contribute makes

82 the next version smarter.

83

84 ****Your data stays yours.**** Using your video to improve the model is ****opt-in**** ? nothing is used to train

85 anything without your explicit permission. We don't publish anyone's footage. ****Sessions involving minors**

86 are never used for model improvement.** The product is built ****private and local-first****: your video, your

87 data, your control.

88

```

89     ---
90
91     ## Where it's going ? toward an **AI assistant coach**
92
93     Today KhelSutra finds rallies and makes reels. The roadmap ? fed by the growing video
94     library above ?
95     turns it into a genuine training companion:
96
97     - **Shot detection** ? recognising smashes, drops, net shots, serves, and clears within
98     each rally.
99     - **Deeper per-rally and per-match analysis** ? patterns, tendencies, what's working and
100     what isn't.
101     - **Per-player & per-match history** ? track a player's progress over weeks and months,
102     not just one clip.
103     - **Personalised insights** ? an **AI assistant coach** that supports (never replaces)
104     the real coach,
105     surfacing what to work on next.
106     - **More sports** ? table tennis next, then other racket/court sports.
107     - **Your language** ? Kannada and Hindi, alongside English.
108
109     > **Read this honestly:** the items in this section are the **direction**, not today's
110     buttons. What you
111     > can use *today* is rally detection + highlight reels + the play-time summary. The
112     deeper analysis arrives
113     > as the video library grows ? which is exactly why your footage and feedback matter so
114     much.
115
116     ---
117
118     ## What we're focused on over the next ~4 weeks
119
120     1. **Growing the library of real academy footage** ? especially busy **multi-court**
121     halls (the hard,
122     valuable case). This is the single biggest lever on quality.
123     2. **Field conversations with academies** ? understanding how coaches and players
124     actually work with
125     video, and what would make this genuinely useful.
126     3. **Recording sessions and showing academies their own footage** as reels ? "this is
127     what it looks
128     like for you."
129     4. **Sharpening multi-court accuracy** and inviting a few more academies into the
130     private trial.
131
132     ---
133
134     ## What KhelSutra is **not** (so expectations stay honest)
135
136     - It's an **early, invite-only trial** ? useful, but still improving; you'll hit rough
137     edges.
138     - It is **not a line-calling, scoring, or umpiring system** ? it finds rallies and
139     highlights.
140     - **Shot detection and the deeper "AI coach" analytics are on the roadmap, not available
141     today.**
142     - It is **not a replacement for a coach** ? it's a tool that saves coaches time and adds
143     insight.
144     - Accuracy depends on **capture quality** ; messy handheld/broadcast footage is a known
145     weak spot.
146     - **Pricing isn't set** ? we're still learning what's fair; today's invited use is about
147     getting it right.
148
149     ---
150
151     ## Quick answers to common questions
152
153     ***"How accurate is it?"***

```

136 Very good on a clean, single-court recording. A busy multi-court hall is harder ? and is exactly what

137 we're improving. The best answer is to *show* a sample reel rather than quote a number.

138

139 ***"Do I need a special camera or hardware?"**

140 No. A phone or GoPro on a tripod, behind the court, whole court in frame. We provide a one-page checklist.

141

142 ***"Is my video private? What happens to it?"**

143 It's built private and local-first ? your video stays yours. We never use anyone's footage to improve the

144 model without explicit opt-in, we don't publish footage, and **footage involving minors is never used**.

145

146 ***"Can it tell me the score / who won / call lines?"**

147 Not today. Today it finds rallies and builds a highlight reel + a play-time summary.

148

149 ***"Does it work for doubles and tournaments?"**

150 Yes for finding rallies. Crowded multi-court tournament halls are the hard case we're specifically working on.

151

152 ***"Will it get better?"**

153 Yes ? that's the whole idea. The more real footage runs through it (with permission) and the more coaches

154 correct it, the sharper it gets, and the sooner shot detection and deeper analysis arrive.

155

156 ***"Which sports?"**

157 Badminton today; table tennis and other racket sports are on the roadmap.

158

159 ***"When can I use it / how do I join?"**

160 It's an invite-only private trial. If you're interested, the best next step is to let us record a session

161 and show you your own court and players as a reel.

162

163 ---

164

165 *KhelSutra · khelsutra.guru · Internal & invited-trial overview, June 2026.*

166 *Field associates: the conversation script and visit rules live in the separate Field Visit Playbook.*

167

4. assistant (2026-06-15T15:00:02.952Z)

I have read the draft and all four lens reviews. Let me verify the two contested factual claims against the repo before finalizing: the watermark casing and the actual data-flow (local vs cloud) to resolve the conflict between the lenses.

5. user (2026-06-15T15:00:04.707Z)

```
config.json:79:      "text": "Generated by Khelsutra.guru",
backend\config\models.py:178:    text: str = "Generated by Khelsutra.guru"
tests\test_watermark.py:25:    assert wm.text == "Generated by Khelsutra.guru"
tests\test_watermark.py:165:                                watermark={"enabled": True, "text":
"Generated by Khelsutra.guru"})
tests\test_watermark.py:195:                                watermark={"enabled": True, "text":
"Generated by Khelsutra.guru"})
```

6. user (2026-06-15T15:00:05.777Z)

```
docs\CODE_MAP.md:99:Distill Gemini->local: calibrate_local (thresholds) OR distill_local
(learned classifier) @backend/eval/{calibrate_local,distill_local}.py
docs\CODE_MAP.md:101:Gemini refinement driver (tuning, standalone):
gemini_refine.{refine_window,full_video_rallies} @backend/eval/gemini_refine.py
```

docs\CODE_MAP.md:113:[Omitted long matching line]
docs\CODE_MAP.md:146:[Omitted long matching line]
docs\CODE_MAP.md:166:- `@backend/eval/calibrate\_local.py` ? distill Gemini->local thresholds (windowing + net-crossing gate, no cloud at runtime). `::local\_grid` (180 cfs),
`::make\_local\_predict\_fn` (adds `rally\_gate.apply\_gate` when `min\_crossings>0`),
`::build\_videos` (? `SystemExit` if a labels file yields no rallies), `::run`/`::main`. \$
`LocalConfig=(velocity,merge,min\_dur,min\_crossings)`, `BASELINE\_LOCAL=(0.02,2.0,2.0,0)`;
manifest JSON shared with distill_local.
docs\CODE_MAP.md:176:[Omitted long matching line]
docs\CODE_MAP.md:177:[Omitted long matching line]
docs\CODE_MAP.md:203:[Omitted long matching line]
docs\CODE_MAP.md:204:[Omitted long matching line]
docs\CODE_MAP.md:282:- **Gemini free-tier / API-key dead path:** hybrids + gemini return `[ ]` (not error) if `{PROVIDER}\_API\_KEY` missing OR WSL healthcheck fails ? a silent no-op.
`analyze\_video` also returns `[ ]` on real failure -> empty != "no rallies". The reporting
`silent\_provider\_noop`/`ai\_empty\_vs\_no\_rallies` rules exist precisely to disambiguate this in the report.
docs\CODE_MAP.md:318:| Distill Gemini -> local (thresholds / learned) |
`@backend/eval/calibrate\_local.py::main` (thresholds) OR `@backend/eval/distill\_local.py::main` (learned classifier); both `--manifest` |
docs\CODE_MAP.md:319:| Refine WASB candidates with Gemini (tuning) |
`@backend/eval/gemini\_refine.py::main` (`--full-video` skips `--trajectory`; needs `GEMINI\_API\_KEY`) |
docs\CODE_MAP.md:336:All pure-logic / mocked-boundary unit tests; no WSL/GPU/Gemini/ffmpeg actually invoked. `@run\_all\_tests.py` (repo root) is the suite runner.
docs\CODE_MAP.md:345:| `test\_gemini.py` | `pipeline/segmenters/gemini.GeminiPlaySegmenter` (single vs chunked path, per-chunk offset, `shot\_count`<`shuttle\_exchange\_count`, chunk re-encode/`ai\_chunk\_scale\_width` args, oversize-skip -> run_signals + ERROR log) |
docs\CODE_MAP.md:375:| `test\_providers.py` / `test\_provider\_gemini.py` | `providers` registry + GeminiProvider lifecycle (`MAX\_UPLOAD\_MB` guard refuses LOUDLY w/ `oversize\_skipped` metric, non-list JSON -> [], cleanup delete) |
docs\COMMERCIALIZATION.md:39:| C4 | **Runtime AI strategy (A/B/C + owned model)** | **P0** | ? | Live = **C) Gemini 2.5 Flash, paid tier**. Long-term = **own a self-hosted generative video-QA model** (fine-tuned open VLM) as primary, **paid Gemini always-ready as fallback** via the provider registry. | Make "provider registry + paid fallback" a core principle; build the owned model **after** the scaffolding/flywheel are solid. See `PLATFORM\_ARCHITECTURE.md` §5A. |
docs\COMMERCIALIZATION.md:43:[Omitted long matching line]
docs\COMMERCIALIZATION.md:45:| C7 | **Hosting unit-economics** | **P2** | ? | Per-video cost is *cheap* for Gemini (~\$0.05 input for a 10-min clip); risk is volume × latency, not unit price. Self-host/offline costs unmeasured. | **Benchmark \$/video & latency on the target VM** for A/B/C before pricing. See §5. |
docs\COMMERCIALIZATION.md:67:| **Gemini API** (paid tier) | service terms | Paid: no training/review | ? LIVE WATCH | Must bill the paid tier to keep user video private (C4). |
docs\COMMERCIALIZATION.md:89:- **C) Keep Gemini 2.5 Flash (paid).** Fastest, cheap per video, video-native;
docs\COMMERCIALIZATION.md:94:OpenAI (no training by default; ZDR available); Gemini remains strongest video-native.
docs\COMMERCIALIZATION.md:122:- **C) Gemini 2.5 Flash (paid):** input \$0.30/1M tokens, output \$2.50/1M; video ~263
docs\COMMERCIALIZATION.md:126: VRAM (L4/A10). \$/video vs Gemini is **open ? measure**.
docs\COMMERCIALIZATION.md:196:4. **Real per-video GPU cost & latency** of self-hosted VLM vs Gemini paid at expected
docs\BACKLOG.md:19:- ? **WSL temp/frame/cache + Gemini remote-file GC** ? staged videos, PNG frame dumps, `\_wasbcache`, and orphaned Gemini uploads are never cleaned. Needs a retention policy. **Revisit:** before any volume/batch processing.
docs\BACKLOG.md:30:- ? **Real GPU \$/video & latency** ? benchmark the three LLM-step options (offline ActionFormer vs self-hosted Qwen2.5-VL-7B vs Gemini paid-tier) on the target VM before pricing the product. **Revisit:** when choosing the production LLM step.
docs\archives\MONETIZATION_AUDIT.md:26:| **Gemini API** (service) | Commercial terms (service, not code) | Paid tier: **no** training / no human review. Free tier: **yes** (trained on + human-reviewed) | ? **WATCH** | **Must bill the API** (paid tier) to keep user video private |
docs\archives\MONETIZATION_AUDIT.md:66:| **C) Keep Gemini** | Gemini 2.5 Flash, **paid tier** | Service terms | Fastest; ~cheap per video; ongoing metered cost + external dependency. Paid tier required for data privacy. |
docs\archives\MONETIZATION_AUDIT.md:79:- **C) Gemini 2.5 Flash (paid):** input **\$0.30 / 1M

tokens** (text/image/video), output **\$2.50 / 1M**. Video tokenizes at **~263 tokens/sec** at default resolution ? a 10-min clip ? 158k input tokens ? **~\$0.05 input** (+ small output). Per-video API cost is **cheap** ; the economic risk is $\text{volume} \times \text{latency}$, not unit price. Flash-Lite (\$0.10/\$0.40 per 1M) is a cheaper tier; Pro is ~10× pricier.

docs\archives\MONETIZATION_AUDIT.md:80:- **(B)** Self-hosted Qwen2.5-VL-7B: **\$0 API**, cost = GPU-hours amortized over throughput. Needs ?24 GB VRAM (L4/A10-class). Real per-video cost/latency vs. Gemini is an **open question** ? measure before committing.

docs\archives\MONETIZATION_AUDIT.md:103:- **Anthropic (Claude):** commercial terms ? **no** training on customer Inputs/Outputs, customer **owns** Outputs, may power your product; bars competing-model use. ? **No native video** (frames must be sent as images) ? weaker than Gemini for this use case.

docs\archives\MONETIZATION_AUDIT.md:105:- **Gemini** remains the strongest **video-native** hosted option; keep it on the **paid tier**.

docs\archives\MONETIZATION_AUDIT.md:113:4. **Real per-video GPU cost & latency** of Qwen2.5-VL-7B (L4/A10) vs. Gemini paid tier at expected throughput.

docs\archives\MONETIZATION_AUDIT.md:121:- Gemini terms/data/pricing: ``ai.google.dev/gemini-api/terms``, ``.../docs/data-usage``, ``.../docs/pricing``.

docs\archives\decisions\DECISIONS.md:10:The pipeline uses an AI multimodal provider (Gemini) for **semantic** rally segmentation.

docs\archives\decisions\DECISIONS.md:17:- **Multimodal LLM APIs** (Gemini, Claude, GPT-class) are the wrong tool for precise

docs\archives\decisions\DECISIONS.md:157:The default high-quality path (``yolo_hybrid``) refines YOLO candidate windows with **Gemini**,

docs\archives\decisions\DECISIONS.md:163:active-frame count, track-id count) stored in ``candidate_segments.stats``. Gemini is only

docs\archives\decisions\DECISIONS.md:164:the Stage-2 refiner. So the rally annotations can **train a classifier to replace Gemini**,

docs\archives\decisions\DECISIONS.md:182: score it with the existing harness against ``motion`` and a Gemini reference.

[Showing results with pagination = limit: 40]

7. assistant (2026-06-15T15:00:09.880Z)

Both contested facts are confirmed. Watermark ships as "Generated by Khelsutra.guru" (lowercase s) and the production path uses paid Gemini cloud AI. Let me verify the privacy-deferral guardrail and the free-tier pooling claim that the consent-safety lens cited, since these drive the most serious BLOCKING edits.

8. user (2026-06-15T15:00:11.950Z)

docs\ANALYZERS_AND_RECONCILERS.md:283:distilled labels or a new (minors-gated) annotation task ? the same gate as feature-level fusion. If you can

docs\CODE_MAP.md:114:[Omitted long matching line]

docs\CODE_MAP.md:282:- **Gemini free-tier / API-key** dead path: **hybrids + gemini** return ``[]`` (not error) if ``{PROVIDER}_API_KEY`` missing OR WSL healthcheck fails ? a silent no-op. ``analyze_video`` also returns ``[]`` on real failure -> empty != "no rallies". The reporting ``silent_provider_noop``/``ai_empty_vs_no_rallies`` rules exist precisely to disambiguate this in the report.

docs\COMMERCIALIZATION.md:43:[Omitted long matching line]

docs\FIELD_VISIT_PLAYBOOK.md:62:students are 18 or older ? "senior batch" players who are minors do NOT count. Then ask 2?3 of them:

docs\FIELD_VISIT_PLAYBOOK.md:107:AND the adult players being recorded; **sessions with minors** are not recorded for our library ? adult sessions only;

docs\HOSTING_PLAN.md:39:- **Upload limit:** free-tier proxy caps request bodies at **100 MB** ? SPA uploads in ?90 MB

docs\HOSTING_PLAN.md:64:- Post-credits estimate: Cloud Run ~\$0?5 (free tier: 2 M req + 180 K vCPU-s/mo, scale-to-zero) .

docs\archives\decisions\DECISIONS.md:245:3. **LLM step** ? keep all three strategies open: (A) eliminate via an offline temporal segmenter (**ActionFormer**, MIT ? the Phase-4 direction), (B) self-host **Qwen2.5-VL-7B/32B** (Apache-2.0) or **InternVL2.5** (MIT), or (C) keep **Gemini 2.5 Flash** on the paid tier (required ? the free tier trains on and human-reviews uploaded video). The ``google-genai`` SDK itself is Apache-2.0 (clear).

docs\I18N_PLAN.md:226:- **No minors / biometrics** scope creep. **i18n** touches presentation only; it introduces no new

docs\MULTI_SIGNAL_FUSION_PLAN.md:203: and the ****paid-LLM/training guardrail + minors-exclusion**** apply to any future person-training data.

docs\PLATFORM_ARCHITECTURE.md:387:consented, minors-excluded video (biometric, \$7); (3) the ****inference plan**** for the size

docs\PLATFORM_ARCHITECTURE.md:446: consent bar (pooled data, biometric/minor care ? \$7); ****minors excluded by default****.

docs\PLATFORM_ARCHITECTURE.md:511: ****not the consumer/free tier****; set short/zero retention, document it in the DPA, and

docs\PLATFORM_ARCHITECTURE.md:518: what merely passes through a tenant's bucket. ****Exclude minors' video from the training**

docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md:29:- ****Exclude minors; per-user training data needs a separate explicit opt-in; split data by match.****

docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md:165:VRAM/token cost · single-model-multisport-temporal generalization unproven · DPA + no-train (minors-excluded) for cloud

docs\PRODUCT_FIELD_ASSOCIATE_FLYER.md:76: ****minors are never recorded**** for our use, period. In regular visits you record

docs\PMF_FIELD_SIGNALS.md:164: minors don't qualify; skip and note it when no adults are available.

docs\PMF_FIELD_SIGNALS.md:166: players; ****sessions with minors are never recorded for our library**** (training-pool

docs\POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md:35: Priority: multi-court badminton (target distro + ship target), then ?3 matches per new sport (TT first; pickleball labels exist but ball detector blocker). Adult sessions only (minors-exclusion). Per video: label (Roora v8 + guidelines) ? curate import-local (Proprietary license) ? re-run harness.

docs\STORAGE_SHARING_MODEL.md:89:~\$0.05/run ? decide whether the free tier includes unlimited processing or N matches/month

docs\STORAGE_SHARING_MODEL.md:95: GPU. A free tier on the owned model is **sustainable at scale**; one on Gemini bleeds.

docs\STORAGE_SHARING_MODEL.md:98: target ? wrapper's 0.66 on multi-court) ? free tier literally improves every quarter ?

docs\STORAGE_SHARING_MODEL.md:100: the free-tier product engine.

docs\STORAGE_SHARING_MODEL.md:101:3. ****Independence & privacy:**** free tier immune to provider 503 waves, price hikes, and terms

docs\STORAGE_SHARING_MODEL.md:144:files), sources auto-expire; ****compute**** ? free tier = owned model (cheap) + N runs/mo cap +

docs\STORAGE_SHARING_MODEL.md:153:Consent-at-capture for other players (Phase-2 recording rules) · ****minors never on public**

docs\STORAGE_SHARING_MODEL.md:172:5. Ratify ****"free never touches Gemini"**** (owned-model-only free tier)?

docs\UI_ALPHA_EXECUTION_PLAN.md:68:`ai\_max\_workers=1`, 6-attempt backoff ? learned 2026-06-10) · CF free-tier 100 MB body cap

docs\VIDEO_LOCALITY_MODEL.md:21:under the cap the file moves as sequential ?90 MB parts through a free-tier edge

docs\README.md:6:[Omitted long matching line]

docs\ROORA_LABELING_GUIDELINES.md:215:> - ****Exclude minors entirely.**** If footage contains children, that video does ****not**** enter the

docs\UI_PROPOSAL.md:55:| B2 | ****Chunked upload endpoint**** | `POST /api/upload` accepting ?90 MB parts (free-tier proxy body limit is 100 MB), assemble ? MD5 ? register under `allowed\_video\_root/<user>/`. Size cap (e.g. 4 GB) + mp4/mov allowlist. |

docs\UI_PROPOSAL.md:78:1. ****Drop 1 ? Accounts & academies (P1):**** real auth (Firebase Auth ? GCP-native, free tier ? or

docs\UI_PROPOSAL.md:89: academy \$300?1,500/yr equivalents, tuned per-match for India), free-tier match allowance;

docs\UI_PROPOSAL.md:91: attestation at upload (guardrails: minors excluded from training pool; separate explicit opt-in).

docs\QUALITY_ITERATIONS.md:120:the baseline batteries); ****free-tier videos ARE pooled**** into the shared baseline (consent baked into free terms,

docs\QUALITY_ITERATIONS.md:121:minors-exclusion enforced at the pooling query); and the previously-pending structural-guard sub-decision is now LOCKED ?

docs\PERSONALIZATION_PLAN.md:28:- ****The owner's contribution-flywheel + free-tier-pooling closes the one real hole**** the counter-case

[Showing results with pagination = limit: 40]

docs\HOSTING_PLAN.md
docs\FIELD_VISIT_PLAYBOOK.md
docs\PRODUCT_FIELD_ASSOCIATE_FLYER.md
docs\STORAGE_SHARING_MODEL.md
docs\PERSONALIZATION_PLAN.md

10. assistant (2026-06-15T15:00:23.804Z)

Let me verify the two most consequential consent-safety claims directly: the privacy-deferral guardrail in PERSONALIZATION_PLAN.md and the HOSTING_PLAN data flow.

11. user (2026-06-15T15:00:25.241Z)

```
135         Recommended sequence: **run a golden-set regression first**, bring the number, *then*
approve.
136     3. **Per-user `user_id` for the alpha** ? device/install UUID acceptable, or must auth
land first?
137     4. **Local-install flywheel default** ? fully-local zero-pooling until consent is
explicitly granted?
138
139     ## 6. Honest risks (carried from the analysis)
140
141     - **Small-N per-user eval noise** is central and *worse* per-user ? mitigated
structurally: the gate
142     refuses below `min_n` and falls back to the baseline; never auto-promotes in the
regime the tooling
143     exists to refuse in. PR-0.5 confirms-or-kills the core premise before any serving
build-out.
144     - **Cold-start** ? the day-0 Gate-A battery (zero per-user labels) is the floor for the
median user.
145     - **Privacy deferral** ? do not advertise "data never leaves your machine" until on-
device ships;
146     near-term pooling carries the full cloud (DPA / consent / minors) posture.
147     - **Compute** ? heavyweight VLM personalization (on-device Qwen3-VL fine-tune) is **out
of v1 scope**;
148     v1 is the cheap LogReg / threshold / cue-ON-OFF layer only.
149
150     ## 7. References
151     - `scratch/PERSONALIZATION_PIVOT_ANALYSIS.md` ? the decision-grade strategy analysis
(pros/cons, harness-fit, counter-case, recommendation).
152     - `scratch/PERSONALIZATION_FEATURE_DESIGN.md` ? the code-grounded design (schema,
serving bridge, APIs, critical files).
153     - Built: `backend/eval/per_user_gate.py` (#141), `backend/eval/per_user_eval.py` (#142).
154     - Harness: [EXPERIMENT_HARNESS.md](EXPERIMENT_HARNESS.md), `backend/eval/experiment.py`,
`backend/eval/eval_stats.py`.
```

12. user (2026-06-15T15:00:25.851Z)

```
1     # Hosting Plan ? khelsutra.guru
2
3     **Status:** PROPOSAL ? awaiting owner ratification.
4     **Date:** 2026-06-10 · **Constraints:** budget $10?25/mo · GCP credits available ·
DigitalOcean
5     acceptable · alpha backend = owner's GPU box behind a tunnel · 10?25 alpha testers ·
registrar = GoDaddy.
6     **Companion:** `docs/UI_PROPOSAL.md`.
7
8     ## 1. Recommendation in one line
9
10    **Alpha: free edge + your box** (Cloudflare DNS/Pages/Access/Tunnel ? FastAPI on the GPU
box) at
11    **$0/mo**, saving GCP credits; **Beta: GCP control plane + your box as pull-worker**
(Cloud Run +
12    GCS + Firebase Auth, credits-covered), keeping the edge where it's free.
```

```

13
14     Rationale for the Cloudflare pieces despite the GCP preference: GCP has no free
equivalent of the
15     Tunnel + Access combo (IAP requires a load balancer, ~$18/mo standing cost), and the
alpha needs
16     exactly three things GCP charges for but Cloudflare gives free: a secure tunnel to a
home box with
17     a custom domain, email-allowlist auth (free ?50 users ? fits 10?25 testers), and static
hosting.
18     Credits are better spent at beta on compute/storage, which is where GCP is excellent.
19
20     ## 2. Alpha topology ($0/mo)
21
22     ```
23     tester browser
24         ? https
25         ?
26     Cloudflare edge (free): DNS + TLS + Access (email-OTP allowlist)
27         ??? app.khelsutra.guru ? Cloudflare Pages (React SPA; auto-deploy from GitHub)
28         ??? api.khelsutra.guru ? Cloudflare Tunnel (outbound-only `cloudflared` service on
the box)
29
30         ?
31         ?
32         FastAPI :8000 on the GPU box (Windows)
33         ? Gemini segmenter path (no GPU in request path)
34         ? async jobs (#16), SQLite WAL, output/ on disk
35     ```
36     - **No inbound ports opened** on the home network; `cloudflared` dials out. Home IP
never exposed.
37     - **Auth:** Cloudflare Access in front of BOTH hostnames; backend trusts the
38     `Cf-Access-Authenticated-User-Email` header (verify the Access JWT for rigor) for per-
user scoping.
39     - **Upload limit:** free-tier proxy caps request bodies at **100 MB** ? SPA uploads in
?90 MB
40     chunks (UI_PROPOSAL B2). Match videos (1?4 GB) flow fine as chunks.
41     - **Box offline = service offline.** Acceptable for alpha; SPA shows a friendly
"processing engine
42     offline" banner (a `/api/health` ping). Beta removes this with the queue.
43     - **Costs:** Pages, Tunnel, Access, DNS = $0. Only the Gemini per-run cost
(~$0.05/match) and
44     domain renewal.
45
46     ## 3. Beta topology (GCP credits; ~$15?20/mo post-credits)
47
48     ```
49     browser ??? Cloudflare edge (free, unchanged) ??? app.khelsutra.guru (Pages)
50                                     ??? api.khelsutra.guru ??? Cloud Run
(FastAPI control plane, scale-to-zero)
51                                     ? jobs
table (Cloud SQL Postgres)
52                                     ? signed
resumable URLs
53                                     ?
54                                     GCS buckets
(uploads / reels, lifecycle ? cold)
55                                     ? pull
jobs, outbound-only
56                                     GPU box = first
`byo_worker` (PLATFORM P3)
57     ```
58
59     - Uploads go **browser ? GCS directly** (resumable signed URLs) ? no proxy body limit,
no home
60     bandwidth for ingest; the box pulls only what it processes (WASB/owned-model jobs),

```

Cloud Run
61 handles Gemini-path jobs entirely in-cloud.
62 - Auth graduates to **Firestore Auth** (free; Google + email sign-in; India-friendly
phone OTP
63 later); Access allowlist retires.
64 - Post-credits estimate: Cloud Run ~\$0.05 (free tier: 2 M req + 180 K vCPU-s/mo, scale-
to-zero) ·
65 GCS 100 GB ? \$2.60 + egress \$0.12/GB (reel downloads ? watch this line) · Cloud SQL
smallest
66 ? \$10 ? **~\$15.20/mo**, inside budget. Egress lever if it grows: serve reels from
Cloudflare R2
67 (zero egress) while keeping everything else GCP.
68
69 **## 4. Alternatives considered**
70

13. assistant (2026-06-15T15:00:34.500Z)

All the contested claims are now verified against the repo. The data flow is unambiguous: alpha sends video off the user's premises through Cloudflare Tunnel to the GPU box and to Gemini's cloud API; beta processes "entirely in-cloud." The privacy-deferral guardrail at PERSONALIZATION_PLAN.md:145 explicitly forbids advertising "data never leaves your machine." The watermark ships lowercase. I have everything I need.

A note on resolving the central conflict between lenses: the **grounding** and **overpromise** lenses treat "private and local-first" as defensible product-philosophy language, while **consent-safety** flags it as a hard guardrail violation. The repo's own PERSONALIZATION_PLAN.md:145 is decisive ? it is an explicit internal rule not to advertise locality until on-device ships, and HOSTING_PLAN.md confirms video leaves the premises today. So I am siding with consent-safety: the two "local-first / your video stays yours" instances are BLOCKING. I keep "private" (access-control, defensible) but remove "local-first."

14. user (2026-06-15T15:01:32.796Z)

Output does not match required schema: root: must have required property 'blocking_edits'

15. assistant (2026-06-15T15:01:36.521Z)

The validation failed because my `improvements` array contains entries with extra keys (`replacement\_note`) and the schema requires each item to have exactly `target`, `replacement`, `why`. Let me resubmit with clean objects.

16. user (2026-06-15T15:02:20.058Z)

Output does not match required schema: root: must have required property 'blocking_edits'

17. assistant (2026-06-15T15:02:24.950Z)

The error says the root must have `blocking\_edits`, yet I am providing it. The likely culprit is that one of my array items is still being rejected, causing the whole object to fail validation and the validator to report the top-level required-property error. Looking more carefully: my earlier failed attempt had a stray `replacement\_note` key; this one does not. The remaining suspect is a length/parsing issue. Let me resubmit with shorter `why` fields and confirm every object has exactly the three required keys.

18. user (2026-06-15T15:03:03.143Z)

Output does not match required schema: root: must have required property 'blocking_edits'

19. assistant (2026-06-15T15:03:07.532Z)

The schema reports the root is missing `blocking\_edits` despite my sending it. This suggests the harness may not be parsing my arguments as an object at all ? possibly the payload size is causing a truncation/parse failure, so the whole input deserializes to an empty/invalid object.

Let me drastically shorten the `why` fields and trim content to reduce payload size, keeping every object to exactly the three required keys.

20. user (2026-06-15T15:03:38.379Z)

Output does not match required schema: root: must have required property 'blocking_edits'

21. assistant (2026-06-15T15:03:42.867Z)

The persistent "missing blocking_edits" error despite my sending it strongly suggests a parse failure of the arguments payload, not a content/key problem. The most likely cause is an unescaped character in one of my long string values breaking the JSON. Let me submit a minimal probe with a single trivial item in each array to confirm the schema shape parses at all.

22. user (2026-06-15T15:03:43.831Z)

Structured output provided successfully

23. assistant (2026-06-15T15:03:48.537Z)

The probe confirms the schema shape is correct and parses fine. The full payload was failing on size/content. The em-dash characters (?) in my replacement strings are the likely culprit when combined with payload length. Let me resubmit the full content but replace em-dashes with " - " (hyphen with spaces) and en-dashes/curly characters with plain ASCII to ensure clean serialization.

24. user (2026-06-15T15:04:25.845Z)

Output does not match required schema: root: must have required property 'blocking_edits'

25. assistant (2026-06-15T15:04:31.269Z)

The probe with tiny strings worked, so the issue is payload size, not the em-dashes. The full payload is too large and gets truncated mid-serialization, dropping the second/third argument. I need to materially shrink the total size. Let me trim every `why` to one terse sentence and condense replacements, keeping all 5 blocking + the highest-value improvements only.

26. assistant (2026-06-15T15:05:25.600Z)

API Error: Overloaded